

The Crew Plugin: Technical Reference

Generated from `useful-claude-add-ons@541b2b70` on 2026-09-22. Verify before trusting.

The low-level guide, for the people who install, configure and extend crew. A companion overview covers the same ground in plain language. Written against crew 0.20.11 on `post-207-checklist`, commit 541b2b70.

Design constraint. A role earns its place only if it buys one of three things: an isolated context window, a restricted tool set, or genuinely independent eyes. Everything else is a file, a command, or the user. Every custom subagent loads the entire `CLAUDE.md` hierarchy at startup, so the roster is kept small and the knowledge is kept in files.

1. Layout

Path	Contents	Count at 0.20.11
<code>plugin/crew/commands/</code>	Slash commands, one Markdown file each with YAML frontmatter.	28
<code>plugin/crew/agents/</code>	Role definitions: <code>tools:</code> , <code>model:</code> , description, procedure.	54
<code>plugin/crew/skills/</code>	Procedures loaded on match; each a directory with <code>SKILL.md</code> and references.	20
<code>plugin/crew/hooks/</code> <code>hooks.json</code> and <code>hooks/scripts/</code>	Hook registrations, each event once per shell flavour, and the scripts they run.	20 registrations across 5 events
<code>plugin/crew/tests/</code>	pytest suites, fixtures, and <code>sabotage.py</code> , which reintroduces known bugs and asserts the suite goes red.	65 modules

Path	Contents	Count at 0.20.11
plugin/crew/evals/	claude plugin eval cases with structural graders; runners in scripts/run-plugin-evals.{sh,ps1}.	5 cases
plugin/crew/CONFIG.md	Every configuration key, its layers and its ratchet rule.	19 sections

2. Roles

The Ladder

Tier	Roles	Tools of note	Add when
0	explorer, qa-reviewer	explorer: Read, Grep, Glob only. qa-reviewer: read-only plus Bash.	Always.
1	security, smoke-author, developer	security: no Write or Edit, has Bash. developer: full write, own context, never reviews its own diff.	Security findings reach review; coverage gaps regress; the PM runs work end to end.
2	dba, docs-writer, browser-tester, analyst, planner, infrastructure-architect, scribe, researcher	planner works from an abstracted brief; secondOpinion.sendsCode is false by design.	Per the table in skills/crew-scaling/SKILL.md; /crew:scale reads .crew/metrics.md and recommends.
3	parallel sessions, worktrees, agent teams		Every involved repository has green smoke.

ROLE_TIERS (hooks/scripts/crew_state.py:1205) holds 13 entries across tiers 0-2. Tier 3 names no role; it governs parallelism, and a config that declares it keeps it.

The PM

Standing teammate named `crew-pm`, reached by `SendMessage` so it keeps its transcript. Tools: `Read`, `Write`, `Edit`, `Bash`, `Grep`, `Glob`, `Agent`, `Skill`. No `SendMessage`, no `ListAgents`, so `resume-by-id` belongs to the caller (`commands/pm.md`). Authority is `report-only | act | autonomous`; the stops in `crew_state.AUTONOMOUS_STOPS` hold at every tier: offboarding a role, deleting a codemap or diagram, rewriting `.crew/metrics.md`, destroying git history. Dispatch cap 3 per pass. Write scope: `.crew/**`, `TOD0.md`, `.work/**`, `docs/diagrams/**`, enforced by the role write guard. Since `agents/pm.md`'s most recent change, the PM re-reads full `crew_state.py` output plus `HEAD`, branch and `git status --porcelain` before every dispatch, and on every return runs `rev-parse, merge-base --is-ancestor, log <pre>..HEAD` and `branch --show-current` against its own checkout, so a backward, sideways or foreign-forward `HEAD` is reported rather than assumed away.

Specialists

`crew_state.SPECIALIST_ROLES` lists 40 roles with no tier, onboarded per repository with `/crew:pm onboard <role>` and justified by the stack (the agent file names the file that proves it). Onboarding a specialist never moves the tier. Groups: languages and frameworks (`python-pro`, `node-developer`, `php-pro`, `rust-engineer`, `react-specialist`, `angular-architect`, `dotnet-core-expert`, `dotnet-framework-4.8-expert`, `sql-pro`, `graphql-architect`, `backend-developer`, `api-designer`); cloud and infrastructure (`terraform-engineer`, `network-engineer`, `platform-engineer`, `microservices-architect`, `legacy-modernizer`, `database-administrator`, `git-workflow-manager`, `workflow-orchestrator`, `multi-agent-coordinator`); Microsoft platform (`windows-infra-admin`, `exchange-online-specialist`, `sharepoint-developer`, `power-automate-specialist`, `powershell-5.1-expert`, `powershell-7-expert`, `powershell-security-hardening`, `ad-security-reviewer`); security and compliance (`penetration-tester`, `compliance-auditor`, `code-reviewer`, `architect-reviewer`, `fintech-engineer`, `payment-integration`); review and writing (`qa-researcher`, `kimu-consult`, `ai-writing-auditor`, `skill-author`, `design-bridge`).

Model Routing

`crew_config.py --models` prints the effective provider, model and family per role, which fallbacks are armed, and whether the self-review guard bars anything. The QA order tries Codex first, then a Copilot model, then Claude; a reviewer of the same family as the diff's author is struck (`qa.order` walked by family, never a fixed list of three names). Config keys `dev.roles.*`, `qa.roles.*`, `dev.fallback`, `qa.fallback`, both open tables so an unlisted role is still settable. The

Agent tool's model override applies per dispatch.

3. Commands

Command	Function	Reads / writes
init	Phased, resumable setup; each phase asks before writing.	.crew/config.json, .crew/verify.json, _verify/, CLAUDE.md template.
onboard	Writes the code map; --refresh <subsystem> re-derives one file against the graph.	.crew/codemap/*.md with DERIVED/JUDGEMENT markers and anchor: commits.
ticket, work, split	Scope; implement, review, gate end to end; split an oversized Jira ticket.	.work/tickets/, tracker via MCP.
review	Hostile QA on the diff: Codex, Copilot or a fresh Claude context, first that probes clean; specialist reviewers by path rule.	.work/review/ <name>/ {diff,prompt,out}.txt; a row in .crew/metrics.md.
verify	Builds the verification map from evidence; --all runs the whole map with no budget; --price measures and writes seconds.	.crew/verify.json (tracked).
gate	Exports, removes and restores branch protection.	Bitbucket or GitHub API; export file.
promote	Promotes to the next environment; refuses without a .work/PROMOTIONS.md row and, if change.requireForProduction, without a change request.	.work/PROMOTIONS.md.
change	Files a change request in SDP, Jira or	Tracker via MCP.

Command	Function	Reads / writes
	.work/changes/; refuses while any template question is unanswered.	
emergency	Stands gates down, opens parallel lanes, records what was skipped.	.crew/incidents/.
debug	Systematic debugging: read, reproduce, isolate, prove, then fix. Adapted from superpowers (MIT).	
pm, roster, model, scale, config, upgrade	Manager conversation; role listing; per-role model set and probe; grow or shrink recommendation; config provenance; schema migration (currently to schema crew_state.SCHEMA_CURRENT = 7).	.crew/config.json, ~/.claude/crew/config.json.
plan, survey, docs, reference, diagram, runbook, handoff	Second-opinion design; gap survey; docs touched by a change; API/feature reference with anchors; Mermaid diagrams with %% Anchors;; runbooks; the handoff note.	docs/, docs/diagrams/, .work/HANDOFF.md.
jira-sync, sdp-sync, obsidian-sync	Two-way ticket sync with the local cache.	MCP servers; Obsidian Kanban file.

4. Hooks and Guards

Event	Matcher	Script	Behaviour
SessionStart	*	pm-brief, platform sync, context resume	Injects crew state and any handoff note; corrects platform.os/shell.
PreToolUse	Bash, PowerShell	crew_guards.py via shell shims	guards.terraformApply, guards.forcePush, guards.adminMerge (block ask allow); guards.prodDatabase, guards.prodServer (none read full) against

Event	Matcher	Script	Behaviour
			<code>production.databases/hosts</code> .
PreToolUse	Write Edit	<code>role_write_guard.py</code> via <code>role-write-guard</code> . {sh,ps1}	<code>guards.roleWrites</code> (off by default, per <code>CONFIG.md §18</code> ; block is the floor once enabled). See section 6.
PreCompact	*	handoff writer	Writes <code>.work/HANDOFF.md</code> before summarisation.
Notification	*	<code>pm_pulse.py</code>	Re-briefs the PM with triggers.
Stop	*	<code>verify-gate.{sh,ps1}</code> , <code>context-watch</code>	Runs the matched rules within <code>verify.stopBudgetSeconds</code> (default 60); blocks on failure; keeps a per-rule record even when a sibling rule in the same pass fails. See section 6.

Registration rules: each event registered once per flavour with `shell: "powershell"` on the Windows entry, because a bare hook command goes to Git Bash on Windows; every `.ps1` hook stands down off Windows as its first executable statement (`if ($env:OS -ne 'Windows_NT') { exit 0 }`), so a host carrying both interpreters never runs a hook twice. Every ratcheted key resolves to the narrower of the repository and machine-global layers; a corrupt layer resolves to the strict value; `change.requireForProduction` ratchets the other way (on wins). A guard that can block ships off and needs a committed must-block/must-allow suite, sabotage-tested.

5. Knowledge Files

Artefact	Location	Freshness rule
Code map	<code>.crew/codemap/</code> , un- ignored by name	An anchor: behind HEAD means re-check with <code>git diff --name-only <anchor>..HEAD -- <paths></code> ; empty output means current. Comparison is <code>sha[:7] == head[:7]</code> . Deliberately carries no <code>##</code> Landmines heading of its own; that judgement lives in the repository's <code>CLAUDE.md</code> and the <code>codemap</code> cites it in place rather than restating it.

Artefact	Location	Freshness rule
Code graph	graphify-out/ graph.json and GRAPH_REPORT.md, tracked	Rebuilt by post-commit and post-checkout hooks via graphify update .; a hand build with --no-viz --code- only writes only one of the pair.
Diagram s	docs/diagrams/*.mmd with %% Anchors:	/crew:diagram refresh rewrites only diagrams whose cited files moved.
Metrics	.crew/metrics.md	One row per review; grouped by ticket column for the BLOCK+FIX rate.
Handoff	.work/HANDOFF.md	Injected at session start until finished or deleted.

6. The Stop Gate and the PM: Crew 0.20.11

The current top entry in CHANGELOG.md's [Unreleased] section, filed against crew 0.20.11, closes three related defects in one pass.

Evidence Retention on a Partial Failure

verify-gate.sh used to exit above the point where it synced the per-rule record, so one failing command discarded every rule that had already passed that turn. The sync now runs first; a failing rule's own outcome is deliberately still not recorded under its own key, because recording it there would orphan the marker the moment the rule is edited to fix it.

The Git Bash python3 Landmine

The gate builds a python3 shim from the proved resolver only when python3 itself is genuinely absent, never shadowing a python or py that is already on PATH; it accepts a native C:\...\python.exe interpreter path, and fails closed with a remediation hint when nothing resolves. context-watch.sh, pm-brief.sh and platform-sync.sh now go through the same strict resolver (crew_py_strict), which requires the probed path to be executable, strips a trailing CR, and normalises drive-letter paths. context-watch.sh specifically: honours stop_hook_active before anything else, fails closed on a stub interpreter without claiming the shared handoff marker (so the PowerShell twin's real budget warning is never suppressed by a false alarm), honours context.enabled: false without needing python at all, and keeps the input's own cwd ahead of

The PM's State Going Stale Mid-Pass

agents/pm.md now re-reads full crew_state.py output plus HEAD, branch and git status --porcelain before every dispatch, and on every return runs rev-parse, merge-base --is-ancestor, log <pre>.HEAD and branch --show-current on its own checkout, so a backward, sideways or foreign-forward HEAD is reported instead of assumed unchanged.

render.sh

Handles flags in any position; exits 1 on an empty diagram directory and 2 on contradictory flags; renders to a temporary file so a failed re-render never overwrites the last good output; prints separate ok / skipped / failed counts. Both render scripts (.sh/.ps1) are now tracked executable.

Every hook and script change in this entry carries a must-block/must-allow case that was sabotage-tested on Linux; the agents/pm.md prose change has no automated test of its own, and the new .ps1 resolver cases skip where no PowerShell host exists.

7. Earlier Changes on the Same Line (0.19.90 to 0.19.96)

Version	Change
0.19.90	The idle rule, in both directions: every role that runs a check runs it in the foreground and never ends its turn waiting on a background task; the PM learns that an idle signal is not proof a role stopped, and escalates on a two-cycle recency check rather than restarting live work.
0.19.91	A behaviour-eval suite: five live tests, each tempting one documented rule, graded by structure rather than by a judge model. The first run caught the PM writing code itself.
0.19.92	The role write guard: a hook that enforces each role's write scope from its own tools: definition. Off by default; the strictest setting is the floor once enabled; corrupt configuration forces the strict setting.

- 0.19.93 The Stop-gate fix that first introduced the per-rule verified record, priced rules, reach declarations (local/network/host) and exit 77 as SKIP, replacing the all-or-nothing marker that section 6 above builds on.
- 0.19.95 T-0003: the developer may commit on the ticket's own branch, and the ticket's scope evidence diffs from the commit the ticket started at rather than from the Stop gate's own marker, which answers a different question.
- 0.19.96 A stray !` in commands/verify.md was read by Claude Code as its own bash-execution syntax, so every /crew:verify invocation died before dispatching anything. Fixed by the correct double-backtick construction; a scan of all 54 command files at the time found no other instance.
- 0.20.2 Every registered .ps1 hook stands down off Windows as its first executable statement, closing a defect where a host with both interpreters silently ran every hook twice.

8. Known Items on the Fix List

- .crew/verify.json has no rule naming test_scope_report.py, test_scope_discipline.py or test_scope_base.py by path, and none whose declared paths cover hooks/scripts/scope_report.py or scope_base.py beyond the whole-suite catch-all rule — a regression in the scope layer alone is not caught until that rule's own trigger fires.
- The four docs/guides/*.html guides (this document among them) have no committed test binding the shipped files to crew-house-style's HTML route beyond the skill's own prose; a hand edit dropping the print rules or the brand stylesheet would not be caught by anything in plugin/crew/tests/.
- Regenerating a guide's .docx from its HTML still needs a page-setup and table-width post-processing pass — LibreOffice's HTML-to-OOXML import emits Letter/A4 and margins inconsistently and writes each table's width as an absolute twip value rather than a percentage of the text column — and that pass lives in no committed script in this repository.
- crew-house-style/SKILL.md's citations into resolve_brand.py (:78-82, :317, :66-67, :298-

317, :301) were hand-verified correct at this commit, but the committed drift test covers only the two citations into `build_report.py` in the same section, so the same kind of line-number rot that has hit other citations in this repository would go unnoticed here.

- `plugin/crew/tests/sabotage.py` reports one mutation, in the frozen-artifact path, that goes STILL GREEN under a full run on this Linux host with no identified skip condition explaining it away as a missing interpreter.

Provenance: counts read from the repository at commit 541b2b70 on branch `post-207-checklist`; the crew 0.20.11 entry is `CHANGELOG.md`'s [Unreleased] section, top entry; the earlier version table is drawn from the same file's history for those version strings. Not verified: any change landed after 541b2b70; `scripts/_test/drift-detection.sh`, which this repository's own `CLAUDE.md` requires by hand before a change to the plugin update path, was not run for this document.